BEFORE YOU READ

Twenty integrals in seven families

The integrals on this page group by the kind of integrand. Each row shows the integrand and its antiderivative; add + C to every one.

L0 Special integrals

L1 Power

L2 **x^n** $n \neq -1$	$x^{n+1} / (n+1)$
L2 **1 / x**	$\ln \lvert x \rvert$

L1 Exponential

L2 **e^x**	e^x
L2 **a^x** $a > 0, a \neq 1$	$a^x / \ln a$
L2 **e^{kx}** $k \neq 0$	e^{kx} / k

L1 Trigonometric

L2 **sin x**	$-\cos x$
L2 **cos x**	$\sin x$
L2 **$\sec^2 x$**	$\tan x$
L2 **$\csc^2 x$**	$-\cot x$
L2 **sec x · tan x**	$\sec x$
L2 **csc x · cot x**	$-\csc x$

L1 Inverse trigonometric

L2 **$1 / (1 + x^2)$**	$\arctan x$
L2 **$1 / \sqrt{1 - x^2}$**	$\arcsin x$
L2 **$1 / (x \cdot \sqrt{x^2 - 1})$**	$\operatorname{arcsec} \lvert x \rvert$

L1 Logarithmic

L2 **ln x**	$x \ln x - x$

L1 Logarithmic pattern

L2 **f′(x) / f(x)**	$\ln \lvert f(x) \rvert$

L1 Trigonometric to logarithm

L2 **tan x**	$\ln \lvert \sec x \rvert$
L2 **cot x**	$\ln \lvert \sin x \rvert$
L2 **sec x**	$\ln \lvert \sec x + \tan x \rvert$
L2 **csc x**	$\ln \lvert \csc x - \cot x \rvert$

TAKEAWAY

Where the logarithms come from

Logarithms appear whenever the integrand has the form f′(x) / f(x): 1/x, tan x, cot x, sec x and csc x all follow that pattern.